

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness: agents autonomously regulate whether to act directly, self-reflect, or consult external knowledge during planning.

<FAST>

direct action

<SLOW>

reflection

<KNOW>

retrieval

Current agent training uses 'flood irrigation'

Blanket knowledge and feedback injection can reduce efficiency and robustness.

Existing training paradigms treat knowledge as always useful—even when the agent already knows what to do.



Trajectory imitation

Gold trajectories teach one behavior mode.

Brittle: bypasses step-level uncertainty.



Trial-and-error

External feedback is injected after failures.

Expensive: learns through repeated mistakes.



Knowledge-augmented

Domain knowledge is continuously supplied.

Wasteful: can over-condition or distract.

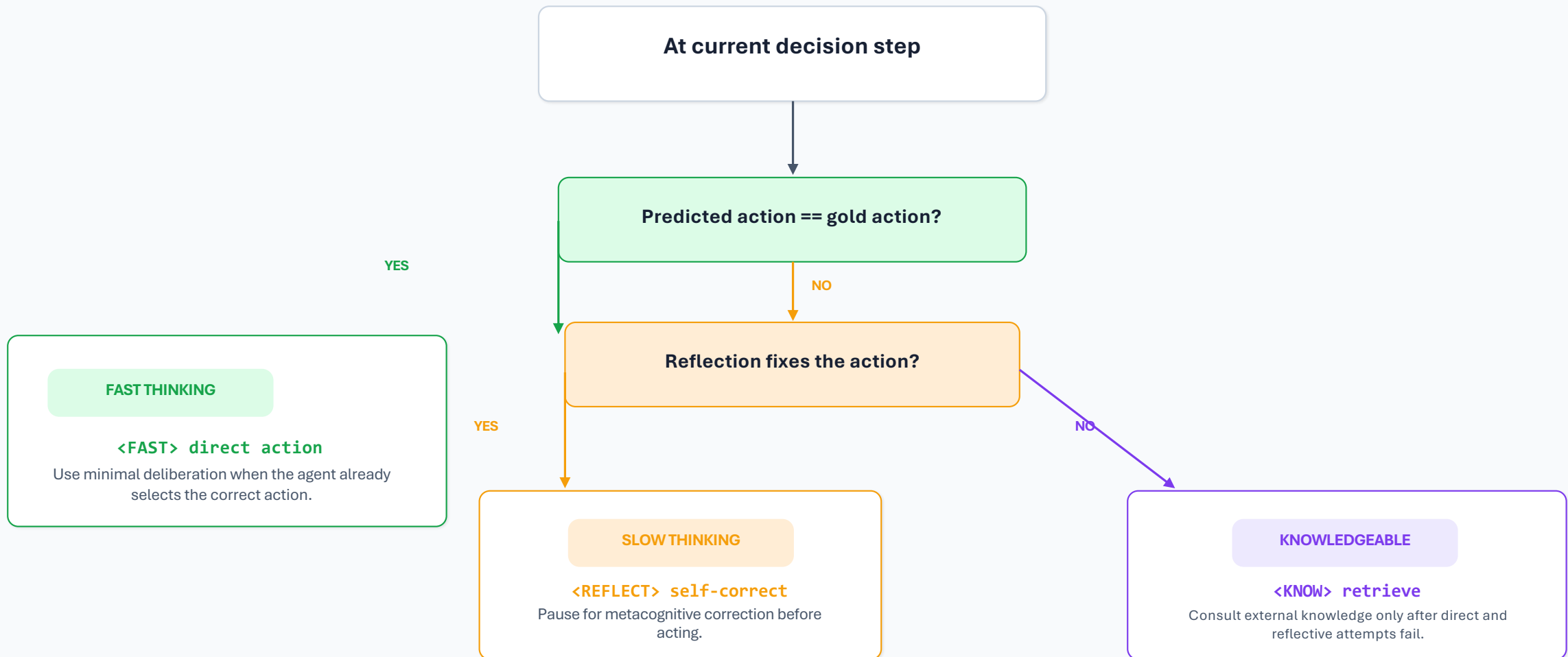
Flood irrigation: inject more knowledge everywhere

Situational self-awareness: ask whether this step needs direct action, reflection, or outside knowledge

KnowSelf reframes agent planning as cognitive-resource allocation, not maximal context injection.

Three situational modes for step-level planning

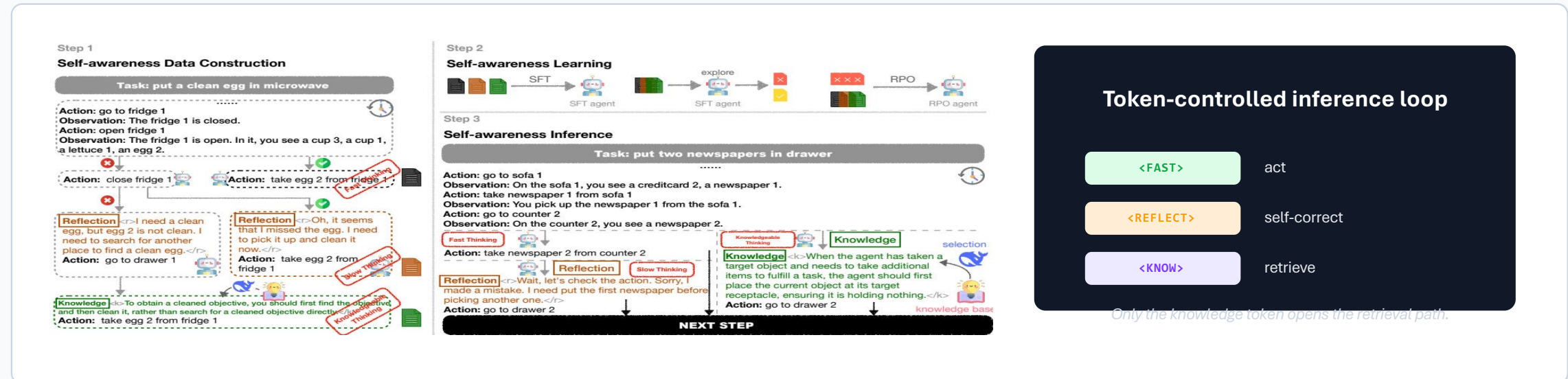
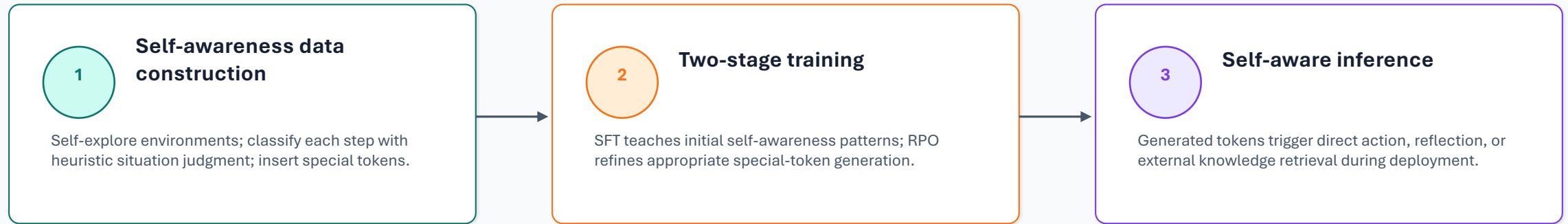
KnowSelf labels each decision by whether the agent can solve it directly, after reflection, or only with knowledge.



Decision criterion is diagnostic, not aspirational: the token marks what the agent actually needs at that step.

KnowSelf pipeline: construct data, train tokens, infer selectively

The full framework converts self-exploration traces into self-aware inference behavior.



Qualitative examples: reflection and knowledge are used only when needed

Two ALFWorld cases illustrate KnowSelf choosing the right cognitive mode where strong baselines fail.

(a) Task: put a hot mug in cabinet	(b) Task: put a saltshaker in drawer
..... Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens. 	 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
..... Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens. 	 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

REFLECTION CASE

“put a hot mug in cabinet”

- OpenAI-O1 opens the microwave, then incorrectly tries to take an egg.
- KnowSelf emits a reflection token and realizes it should heat the mug using the open microwave.

KNOWLEDGE CASE

“put a saltshaker in drawer”

- DeepSeek-R1 gets stuck trying to close an already open drawer first.
- KnowSelf uses a knowledge token: place the object without removing unrelated items.

The examples are not “more thinking everywhere”; they are targeted cognitive interventions.

ALFWorld results: KnowSelf reaches 84.33% with 15.01% knowledge

Success rates (%) across task types; Know% is the proportion of steps using external knowledge.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

What the table shows

84.33%

Llama-8B KnowSelf All

15.01%

knowledge steps

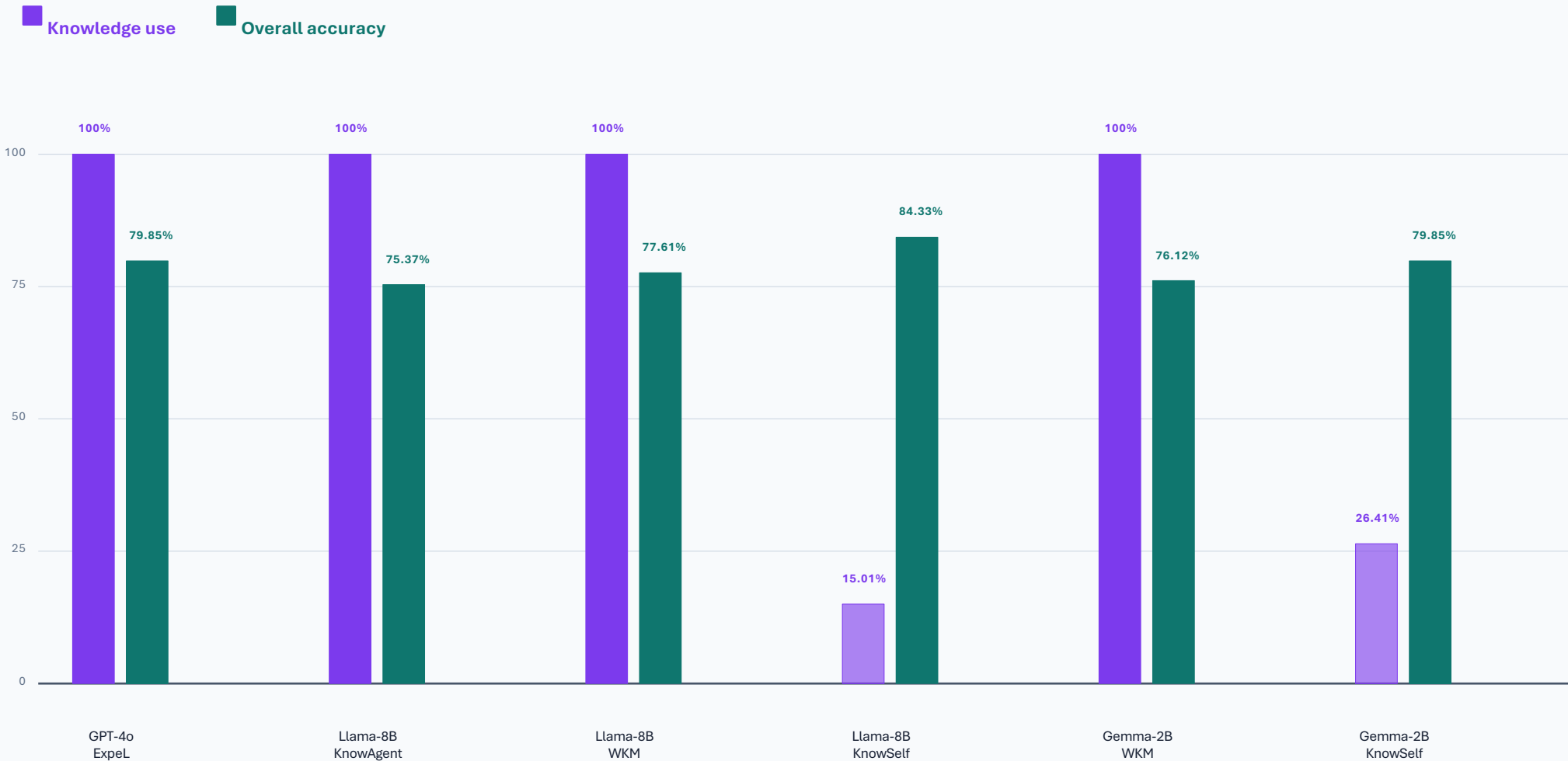
79.85%

Gemma-2B KnowSelf All

KnowSelf outperforms Llama-8B full-knowledge baselines while querying external knowledge on a small minority of steps.

Efficiency advantage: selective knowledge beats full-knowledge defaults

Knowledge usage percentages are compared against overall ALFWorld accuracy across methods and model scales.



Efficiency signal

15–26%

KnowSelf knowledge use

100%

ExpeL / KnowAgent / WKM
knowledge use

*Selective retrieval preserves
accuracy while cutting retrieval
exposure.*

Key takeaways

Self-awareness is the missing control layer for agent planning.

1

Indiscriminate knowledge injection is wasteful and can be suboptimal.

2

Special tokens let agents learn situational self-awareness from lightweight heuristic labels.

3

Llama-8B KnowSelf beats GPT-4o ReAct and GPT-4o ExpeL: 84.33% vs 76.12% and 79.85%.

4

Only 15–26% knowledge usage is needed versus 100% for competing knowledge-heavy methods.

5

The paradigm scales down: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions

- Richer token taxonomies for planning uncertainty.
- Better retrieval policies for dynamic knowledge bases.
- Cross-domain validation beyond ALFWorld.
- Calibration: when should an agent distrust its own reflection?

Bottom line

The strongest agent is not the one that always thinks more—it is the one that knows what kind of thinking the current step requires.